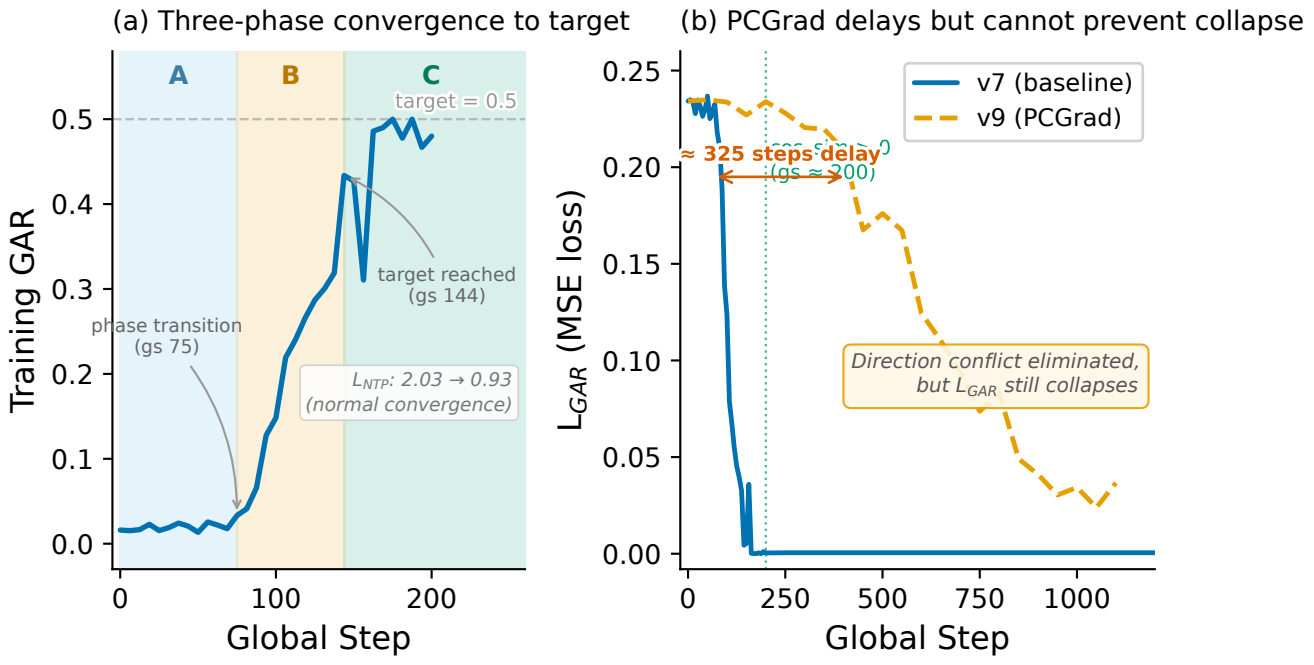
(a) Three-phase convergence to target
A
B
C
target = 0.5
target reached
(gs 144)
phase transition
(gs 75)
L_{NTP}: 2.03 → 0.93
(normal convergence)
Training GAR
Global Step
(b) PCGrad delays but cannot prevent collapse
v7 (baseline)
v9 (PCGrad)
≈ 325 steps delay
(gs ≈ 200)
Direction conflict eliminated,
but L_{GAR} still collapses
L_{GAR} (MSE loss)
Global Step